Transformers

Part II – Introduction to Statistical Learning

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Status: STUB

This chapter has not been fully authored yet. It is scaffolded so the folder/file layout matches every other chapter in the repo. See `README.md` in this folder for how to contribute a full chapter.

Source slide deck: ***`source_slides/Chapter_11__Transformers.pdf`*** *(preserved unmodified from the original repository).*

Editorial note: NUMBERING DRIFT: course chapter 12 by topic order; original file is named 'Chapter_11'. See AUDIT.md.

Planned topics

- Sequence-to-sequence models
- Attention mechanism & the bottleneck problem
- Categories of attention
- Self-attention
- Multi-head attention
- Encoder / decoder architecture
- Positional encoding

TODO

- Expand each topic above into full prose sections with worked examples.
- Add figures under `figures/`.
- Fill in `exercises/python`, `exercises/r`, `exercises/rust` (plain exercise, Rustlings-style buggy exercise, and solution) per the pattern in Chapter 1/2 of Part I.